

Lesson 11:

1. Designing Visuals

- **Visualization Best Practices**
 - Focus on clarity and effectiveness when presenting data.
 - Emphasize the use of appropriate visual encodings.
 - **Edward Tufte's Visual Encoding**
 - A guide for translating data into meaningful visuals.
 - **Conversion of Data into Visualizations**
 - Visuals help draw insights from raw data.
 - Important to use proper encoding to effectively communicate the data.
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2. Data Types

Understanding the different types of data is crucial for choosing the right visualization approach.

- **Nominal Data:**
 - Qualitative; represents categories (e.g., Gender, Blood Group).
 - Can only be compared using equality or inequality (e.g., $=$, $\neq$).
 - **Ordinal Data:**
 - Data from an ordered set (e.g., Customer Reviews).
 - Allows comparisons like $=$, $\neq$, $>$, $<$.
 - **Interval Data:**
 - Fixed interval sets (e.g., IQ level, temperature in Celsius).
 - Comparisons with $=$, $\neq$, $>$, $<$, and subtraction are meaningful.
 - **Ratio Data:**
 - All arithmetic operations are meaningful (e.g., Revenue, Sales).
 - Supports comparisons and operations like $=$, $\neq$, $>$, $<$, $+$, $-$, $\%$.
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3. Effectiveness of Visual Encodings

How humans decode graphs and patterns plays a key role in understanding visualizations.

- **Three Visual Operations** (Cleveland, 1994):
 1. **Detection:** Recognizing patterns or values.
 2. **Assembly:** Grouping geometric objects to understand relationships.
 3. **Estimation:** Assessing magnitude or comparisons.
 - **Gestalt Laws:** Help in grouping objects effectively, such as the Law of Continuity for linking related data points.
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4. Types of Visual Encodings

Different encoding techniques are used for visualizing data, with varying levels of effectiveness.

- **Position on Common Scale:** Most effective for all types of comparisons (Discrimination, Ranking, Estimation).
 - **Position on Identical but Non-Aligned Scales:** Effective, but less so than a common scale.
 - **Length:** Very effective when comparing against a common axis.
 - **Angle:** Useful for ranking or estimation, especially with changes over time.
 - **Volume / Density / Saturation:** Useful for all types of comparison but less effective than other methods.
 - **Color:** Best for discrimination, not suitable for ranking or estimation beyond simple categories.
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5. Data Encoding Techniques

Different encoding strategies for presenting data visually:

- **Color:**
 - Use to distinguish data, but avoid more than six colors in a single chart.
 - Can be effective for discrimination but not for ranking or estimation.
 - **Volume:**
 - Effective for discrimination and ranking.
 - Allows comparison and estimation of differences.
 - **Angle:**
 - Best used to indicate rate of change rather than absolute values.
 - **Length and Position:**
 - Very effective for visualizing differences, rankings, and comparisons.
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6. Expressiveness vs Effectiveness

- **Expressiveness:** How well the visualization shows all facts in the data.
 - **Effectiveness:** How easily the viewer perceives the data through the visual representation.
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7. References

- **Books for Further Reading:**
 - *Data Visualization – Storytelling Using Data* by Sharad Sringswara, Purvi Tiwari, U. Dinesh Kumar.
 - *Storytelling with Data* by Cole Nussbaumer Knaflic.
 - *Information Dashboard Design* by Stephen Few.
 - *Data Visualization: A Successful Design Process* by Andy Kirk.
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